

Practical 6

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In [1]: # Import necessary Libraries
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, recall_score, precision_score, f1_score

# Load the Iris dataset
iris = load_iris()
X_clf, y_clf = iris.data, iris.target

# Split the data
X_clf_train, X_clf_test, y_clf_train, y_clf_test = train_test_split(X_clf, y_clf, test_size=0.3, random_state=42)

# Initialize and train Logistic Regression model
log_reg = LogisticRegression(max_iter=200, solver='lbfgs')
log_reg.fit(X_clf_train, y_clf_train)

# Predict and evaluate
y_clf_pred = log_reg.predict(X_clf_test)
accuracy = accuracy_score(y_clf_test, y_clf_pred)
conf_matrix = confusion_matrix(y_clf_test, y_clf_pred)
recall = recall_score(y_clf_test, y_clf_pred, average='macro')
precision = precision_score(y_clf_test, y_clf_pred, average='macro')
f1 = f1_score(y_clf_test, y_clf_pred, average='macro')

# Output results
print(f"Logistic Regression - Accuracy: {accuracy}")
print(f"Confusion Matrix:\n{conf_matrix}")
print(f"Recall: {recall}, Precision: {precision}, F1 Score: {f1}")
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Logistic Regression - Accuracy: 1.0

Confusion Matrix:

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[[19  0  0]
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 [ 0 13  0]
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```
 [ 0  0 13]]
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Recall: 1.0, Precision: 1.0, F1 Score: 1.0